

CARDIFF METROPOLITAN UNIVERSITY • ICBT CAMPUS • CSE6035

MindCare

Emotion Recognition & Mental Support System Using AI Voice Assistant

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Introduction & Problem Statement

MENTAL HEALTH CRISIS & CLINICAL DEFICIT

- **1 in 4 individuals** globally affected by mental health disorders (WHO, 2022).
- **76% in LMICs** receive zero professional treatment or support.
- **Sri Lanka:** 14.6 suicides per 100k; only **0.2 psychiatrists per 100k**.
- Deep cultural stigma suppresses traditional in-person help-seeking.

CORE TECHNICAL & USABILITY PROBLEM

- Existing digital tools are **monolingual (English)** and text-heavy.
- Typing emotional descriptions imposes **cognitive friction during panic**.
- Traditional chatbots lack **real-time non-verbal affective awareness**.

Proposed Solution: A low-barrier, trilingual (EN / SI / TA) wellness companion integrating real-time facial and speech emotion recognition for non-clinical guided support.

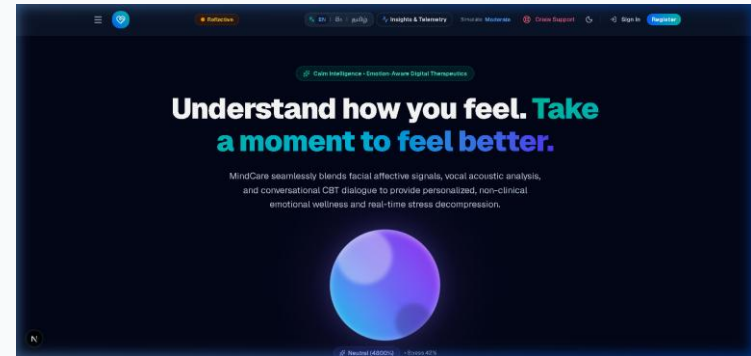


Figure 1: MindCare Trilingual Landing Interface (EVID-SS-001)

Literature Review & Research Gap

01 • UNIMODAL SENSING

Environmental Sensitivity

FER is sensitive to **lighting & posture** (Dinges et al., 2005); SER degrades under **ambient noise**; text increases **friction during panic** (Suri et al., 2026).

02 • SENSOR FUSION

Multimodal Fusion

Combining visual and vocal channels enhances stability (Zhang et al., 2022); **Late Fusion** provides modularity and missing-sensor resilience.

03 • AI GOVERNANCE

Conversational AI Safety

Unrestricted LLMs carry **hallucination and sycophancy risks** in mental health; requires **constrained graph workflows** (Patel et al., 2026; Voultsiou & Moussiades, 2026).

04 • LOCALIZATION

Multilingual Accessibility

Over **90% of affective systems are English-centric** (Zhu, 2023), causing performance degradation and exclusion in South Asian (Sinhala/Tamil) contexts.

THE IDENTIFIED RESEARCH GAP

A critical lack of accessible, low-barrier systems that integrate **multimodal affective sensing, native trilingual support, and constrained conversational safety** within a unified non-clinical wellness platform.

Multimodal Sensing (Face + Voice)

Trilingual Support (EN / SI / TA)

Constrained CBT State Machine

Privacy-Conscious Processing

Aim and Research Objectives

PRIMARY PROJECT AIM

"Develop a trilingual, multimodal and privacy-conscious AI-based wellness companion capable of estimating emotional states and providing guided non-clinical support."

SPECIFIC RESEARCH OBJECTIVES

01

Facial Emotion Recognition

Real-time visual expression classification across 7 universal affective states via CNN.

02

Speech Emotion Recognition

Acoustic emotion estimation extracted from 40-dimensional MFCC audio features.

03

Multimodal Fusion & Stress

Asynchronous late fusion with polar conflict detection and stress mapping.

04

Stage-Aware Guided Support

Constrained conversational therapy using a CBT state machine and deterministic fallback rules.

05

Trilingual Accessibility

Complete interface and speech interaction in English, Sinhala, and Tamil.

06

Privacy-Conscious Processing

Temporary in-memory media processing with client-side sensor consent controls.

Methodology & System Architecture

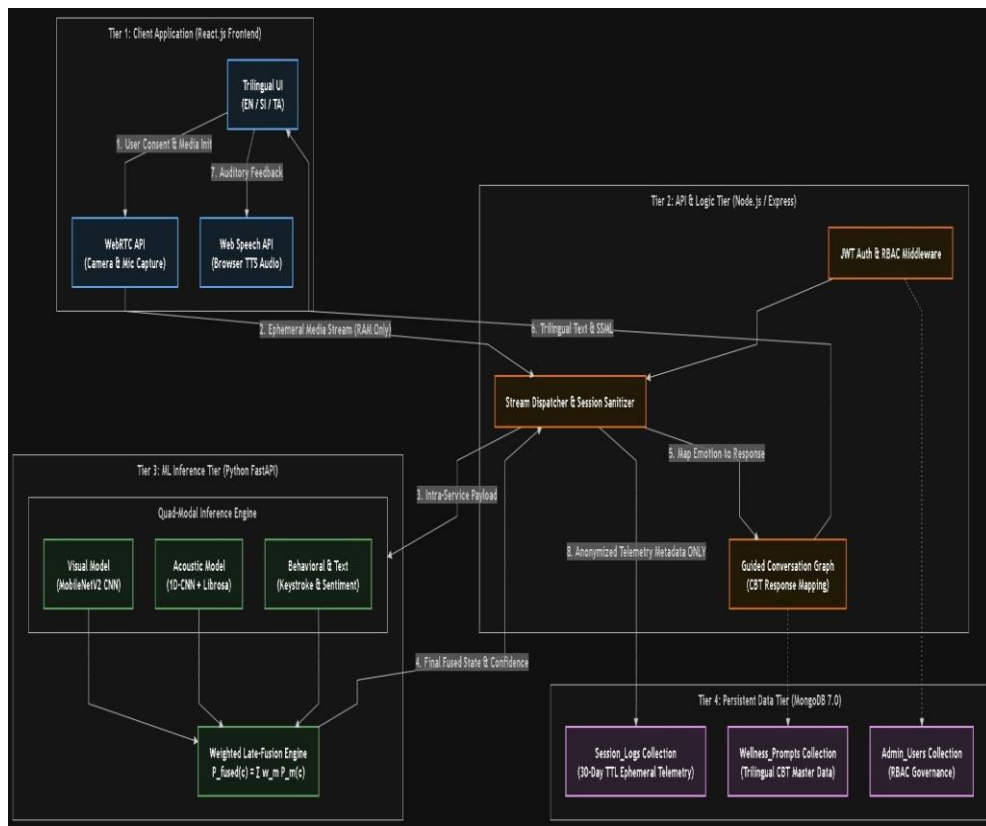


Figure 4: High-Level Decoupled Architecture (Thesis Chapter 5)

TIER 1 • PRESENTATION CLIENT

Next.js & React (TypeScript): WebRTC sensor capture, dynamic trilingual UI (EN/SI/TA), real-time canvas visualization.

TIER 2 • API & ORCHESTRATION

Python FastAPI (Uvicorn ASGI): Full-duplex WebSockets, JWT authentication, non-blocking asynchronous event loop.

TIER 3 • ML & THERAPY ENGINES

TensorFlow & Librosa: Facial 2D-CNN, Speech DNN, 50/50 late fusion, Gemini Flash 3.7 constrained dialogue.

TIER 4 • DATA PERSISTENCE

SQLite 3 via SQLAlchemy ORM: User-isolated session metadata, 3-tier RBAC accounts, and cascade deletion.

Methodology: Adapted 12-Week Agile-Scrum Sprints + CRISP-DM

End-to-End System Workflow & Sequence

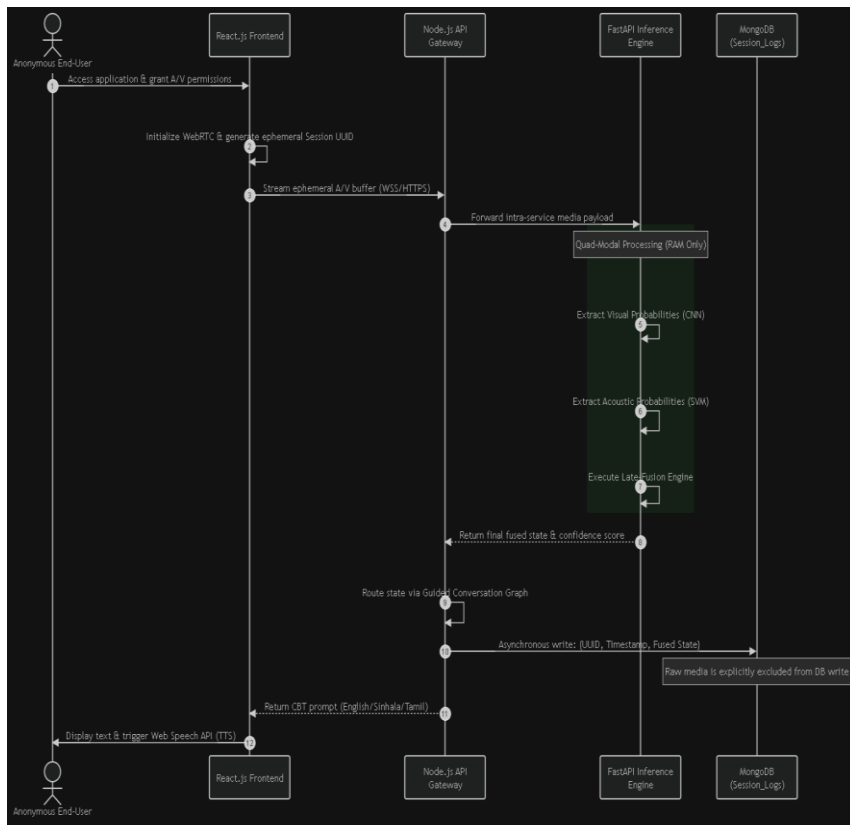


Figure 10: Multimodal Asynchronous Sequence Flow (Thesis Chapter 5)

WebRTC Media Acquisition

Browser captures webcam video frames & audio chunks under user consent.

WebSocket Streaming

Bi-directional full-duplex TCP stream delivers payloads with <50ms network latency.

Parallel Worker Inference

FastAPI dispatches Facial CNN & Speech DNN via non-blocking worker threads.

Late Fusion & Conflict Resolution

Calculates 50/50 weighted state and checks for polar emotion conflicts.

Constrained CBT Dialogue & TTS

Gemini LLM generates structured therapy reply; audio is synthesized via TTS.

Non-Blocking Concurrency: isolates heavy tensor matrix multiplications, preserving a smooth 60 FPS client experience.

ML Implementation & Empirical Results

FACIAL EMOTION • FER-2013 (TEST N=7,178)

61.90%

Baseline 2D-CNN (Production)

FACIAL EMOTION • TRANSFER LEARNING

40.53%

Fine-Tuned MobileNetV2

SPEECH EMOTION • RAVDESS / TESS

53.60%

Baseline DNN (40 MFCCs)

FER-2013 MobileNetV2 Confusion Matrix (Normalized)

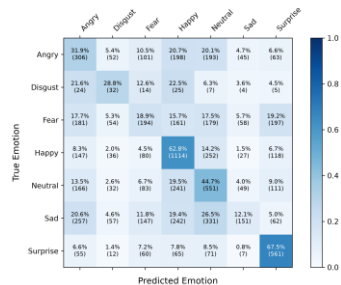


Figure 2: Facial Confusion Matrix — FER-2013 (EVID-SS-151)

Voice Emotion Confusion Matrix (Normalized)

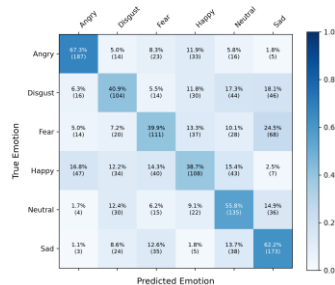


Figure 3: Voice Confusion Matrix — 40 MFCCs (EVID-SS-153)

Empirical Finding: Baseline 2D-CNN outperformed MobileNetV2 in the observed evaluation, indicating a domain gap between pretrained ImageNet features and low-resolution (48×48) grayscale FER-2013 data.

Multimodal Fusion & Stress Estimation

LATE FUSION STRATEGY & CONFLICT LOGIC

$$\text{fused_prob}[e] = 0.50 \times P_{\text{face}}[e] + 0.50 \times P_{\text{voice}}[e]$$

- **Polar Conflict Detection:** Contradictory valences (e.g., Happy Face + Sad Voice) resolve to *"Uncertain"*.
- **Graceful Degradation:** Transparent single-modality passthrough when camera or mic is disabled.
- **Text-Only Fallback:** Default baseline state when sensors are absent.

SOMATIC STRESS INDEX WEIGHT MAPPING

Fear
0.90

Angry
0.85

Sad
0.75

Disgust
0.65

Uncertain
0.50

Surprise
0.45

Neutral
0.30

Happy
0.15

Tri-Tier Stress Levels: Low (<0.35) | Moderate (0.35–0.69) | High (≥0.70)

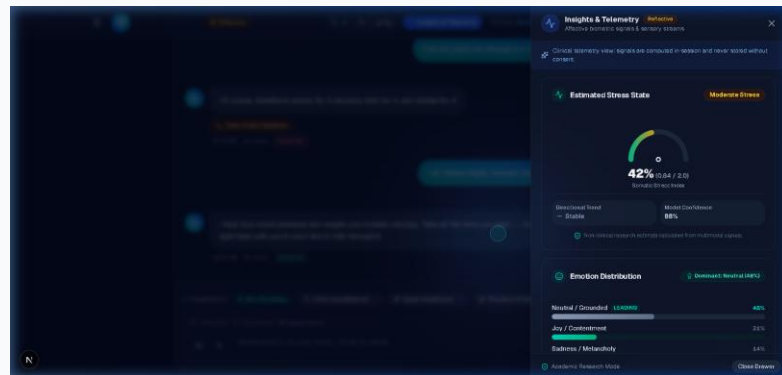


Figure 4: Real-Time Somatic Stress Gauge (EVID-SS-050)

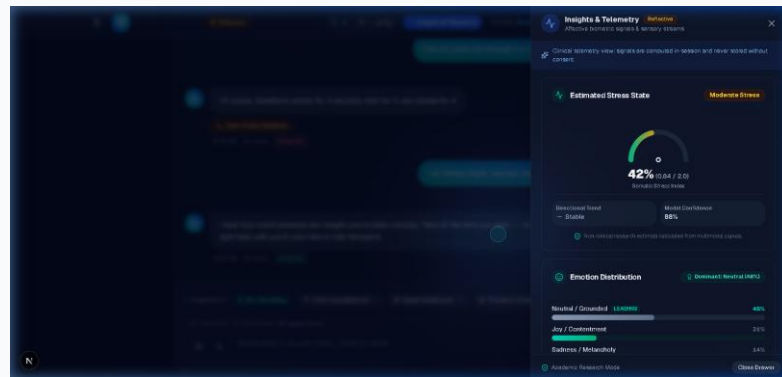


Figure 5: 7-Class Emotion Probability Telemetry (EVID-SS-032)

Therapy, Multilingual Support & Privacy

1. CONSTRAINED CBT THERAPY ENGINE

- **Empathy-First Rule:** Validates feelings before suggesting exercises.
- **5-Stage State Machine:** Crisis, Coping, Early Listening, Exploration, De-escalation.
- **Deterministic Local Fallback:** Local rule engine if LLM exceeds 1.5s.
- **Strictly Non-Clinical:** Guided wellness decompression only.

2. TRILINGUAL CULTURAL LOCALIZATION

- Full native UI scripts & dialogue in **English, Sinhala, and Tamil.**
- **TTS Note:** Browser-native TTS; regional Sinhala/Tamil voice availability depends on host browser/OS.

3. PRIVACY-CONSCIOUS ARCHITECTURE

- **Temporary In-Memory Processing:** Zero server-side raw video saving.
- **Local Privacy Masking:** Client-side Gaussian blur for camera feed.
- **User-Scoped Storage:** Isolated SQLite chat records with GDPR purge.

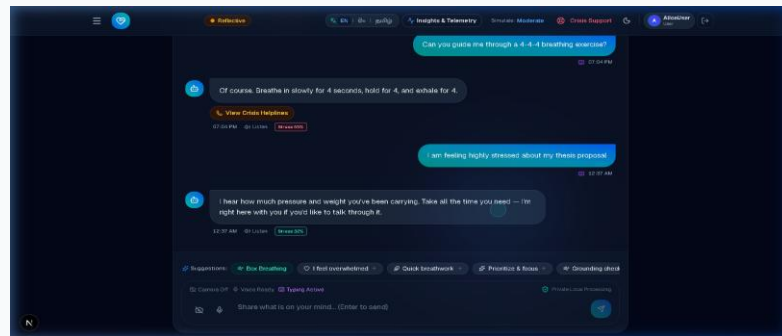


Figure 6: Guided CBT Therapy Workspace (EVID-SS-003)

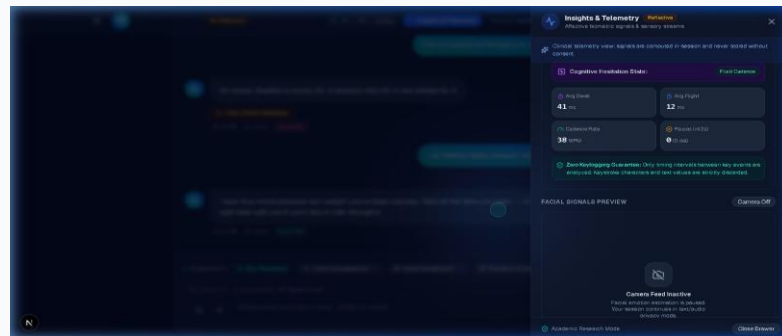


Figure 7: Privacy Settings & Sensor Permissions (EVID-SS-130)

Testing & Empirical Validation

13 / 13 AUTOMATED TESTS PASSED

100% Pass Rate across Integration & Security Test Suite

VERIFIED SYSTEM CAPABILITIES

- **Authentication & JWT:** 24h HS256 lifecycle & identity resolution (/auth/me).
- **3-Tier RBAC Guards:** User, Admin, Super Admin 403 authorization blocks.
- **Database Behaviour:** SQLite schema integrity & cascading deletion.
- **Chat Isolation:** User-scoped queries preventing cross-tenant leakage.
- **WebSocket Reliability:** Heartbeat monitor & non-blocking queues.
- **ML Health Checks:** Facial, vocal & tabular weights loaded successfully.

Rigorous Distinction: Automated tests validate system integration; ML accuracy is verified on held-out test partitions (FER-2013 & RAVDESS).

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MindCare Backend Automated Test Suite Execution (13/13 Passed)

[TherapyBot] gemini-3.7-flash initialized successfully (Therapy & Multimodal STT).
SUCCESS: Face emotion model loaded successfully from best_face_model_v2.h5.
SUCCESS: Voice emotion model loaded successfully from best_voice_model_v2.h5.
SUCCESS: Voice MFCC scaler loaded successfully from voice_scaler_v2.pkl.
SUCCESS: Health/stress model loaded successfully from best_health_model_v2.pkl.

Starting MindCare 3-Tier RBAC & DB Test Suite
=====
[1] Database tables initialized.
[AUTH] New user registered: 'AliceUser' (alice@example.com) with role 'user'
[2] Standard user registered: ID=1 (role=user)
[AUTH] New user registered: 'RootSuperAdmin' (root@example.com) with role 'super_admin'
[3] Super Admin bootstrapped: ID=2 (role=super_admin)
[AUTH] New user registered: 'BobAdmin' (bob@example.com) with role 'admin'
[4] Admin registered: ID=3 (role=admin)
[5] Unauthorized super_admin escalation blocked (HTTP 403).
[AUTH] User login successful: 'AliceUser' (alice@example.com) [user]
[AUTH] User login successful: 'BobAdmin' (bob@example.com) [admin]
[AUTH] User login successful: 'RootSuperAdmin' (root@example.com) [super_admin]
[6] JWT tokens generated for User, Admin, and Super Admin.
[7] /api/v1/auth/me resolved correct identity for all 3 tiers.
[8] RBAC Guard verified on /api/v1/auth/admin/users (User=403, Admin=200, SuperAdmin=200).
[9] RBAC Guard verified on POST /api/v1/admin/create-admin (User=403, Admin=403, SuperAdmin=201).
[10] RBAC Guard verified on GET /api/v1/super-admin/system-stats (Total users=4, stats={'user': 1, 'admin': 2, 'super_admin': 1}).
[11] ChatLog telemetry turn logged and verified for user via ORM.
[CHAT-PERSISTENCE] Saved turn #2 for user 'AliceUser' (Emotion: sadness, Stress: 0.65)
[12] POST /api/v1/chat/save persisted turn #2 under user 'AliceUser'.
[13] GET /api/v1/chat/history verified user isolation (User turns: 2, Admin turns: 0).

ALL 13 RBAC, CHAT-PERSISTENCE, & DATABASE TESTS PASSED WITH 100% SUCCESS!
=====
```

Figure 8: Automated Test Suite Terminal Output (EVID-SS-140)

Live Artefact Demonstration

DEMONSTRATION EXECUTION FLOW

Backend Services: FastAPI & ML
Active

Camera & Mic: WebRTC & Privacy
Blur

Speech Emotion: 40 MFCC Classifier

Language Toggle: EN / SI / TA
i18n

De-escalation: Box Breathing Overlay

Application Login: JWT Auth & 3-Tier
Roles

Facial Emotion: 7-Class Visual ML

Multimodal Fusion: Stress Gauge

Guided Support: Empathy-First
CBT

Admin Governance: RBAC & Logs

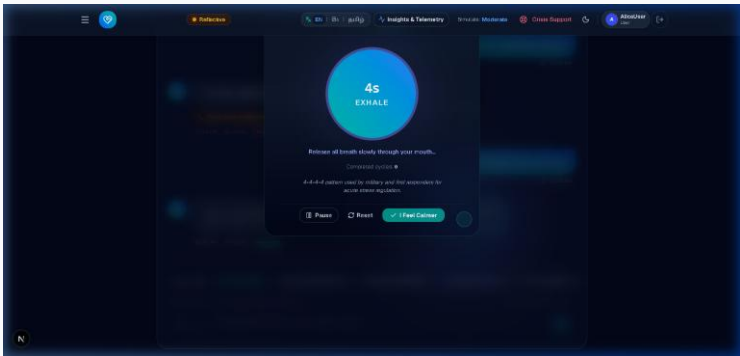


Figure 9: Box Breathing De-escalation (EVID-SS-080)

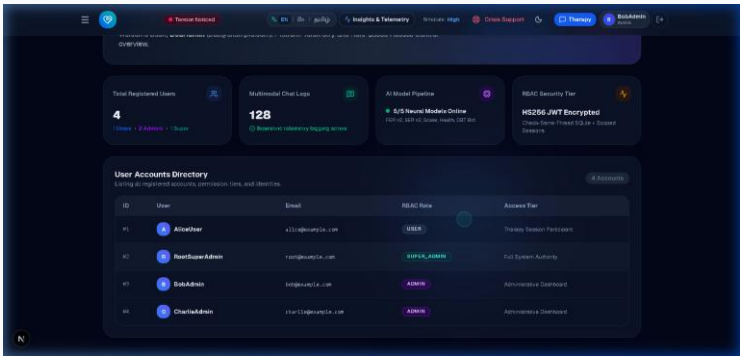


Figure 10: Admin Panel & Governance (EVID-SS-120)

Critical Evaluation: System Limitations

01 • Visual Domain Gap (FER-

2013) contains **low-resolution (48×48) grayscale** crops with ~65% human agreement ceiling. MobileNetV2 underperformed baseline CNN (40.53% vs 61.90%) due to ImageNet feature mismatch.

02 • Platform Dependency

(TTS) browser-native neural voice synthesis in **Sinhala and Tamil relies on OS-level voice pack availability**, requiring cloud fallbacks on unsupported client environments.

03 • Environmental

Sensitivity Speech recognition is susceptible to **background ambient noise** and microphone hardware; facial classification degrades under poor illumination and head rotation.

04 • Evaluation Constraint &

Scope No single unified multimodal benchmark metric exists because models were evaluated independently; system is **strictly non-clinical** without formal longitudinal trials.

Academic Integrity Note: Empirical limitations are presented transparently as engineering findings rather than failures; no unified multimodal accuracy is claimed without paired benchmark verification.

Future Work & Strategic Roadmap

PHASE 1 • DATA & TEMPORAL MODELS

Paired Datasets & Transformers

Train on synchronized audio-visual corpora (e.g. **CREMA-D**); deploy **Vision Transformers (ViT)** for FER and **wav2vec 2.0 / LSTM** for temporal speech dynamics.

PHASE 2 • ADVANCED FUSION

Attention-Based Dynamic Fusion

Replace static 50/50 probability averaging with **learned cross-attention mechanisms** that dynamically weight modalities based on acoustic SNR and face visibility.

PHASE 3 • PHYSIOLOGICAL SENSING

Wearable Biometric Telemetry

Integrate Bluetooth smart bands for non-invasive **Heart Rate Variability (HRV)** and **Electrodermal Activity (EDA)** to enhance objective stress profiling.

PHASE 4 • CLINICAL VALIDATION

Longitudinal Clinical Evaluation

Conduct an **IRB-approved multi-user study** in collaboration with certified psychiatrists to validate longitudinal therapeutic efficacy in Sri Lankan clinical settings.

Strategic Vision: Evolving MindCare from a non-clinical affective companion into an evidence-based digital therapeutic platform for low-resource global mental health.

Conclusion

KEY PROJECT OUTCOMES

- **Trilingual & Multimodal Companion:** Developed an accessible, privacy-conscious AI-based wellness platform.
- **Affective Sensing:** Integrated facial and speech emotion recognition to estimate emotional states.
- **Multimodal Late Fusion:** Combined visual and acoustic probability distributions with conflict logic.
- **Constrained CBT Guidance:** Implemented a 5-stage CBT state machine with deterministic local fallback.
- **Cultural Localization:** Delivered English, Sinhala, and Tamil accessibility with speech interaction.
- **Privacy-Conscious Architecture:** Enforced in-memory media processing and user consent controls.
- **Rigorous Verification:** Validated core system functionality through **13 / 13 automated tests passed**.

KEY FINDING

The empirical evaluation demonstrated the feasibility of combining multimodal emotion recognition with guided non-clinical wellness support, while also identifying limitations in individual model performance, environmental robustness and multimodal benchmark evaluation.

FINAL TAKEAWAY

MindCare demonstrates a practical foundation for an accessible, multilingual and safety-conscious AI-assisted wellness companion.

Thank You

Questions & Discussion

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Key Academic References (Thesis Chapter 2):

- **Zhang et al. (2022)**. Multimodal affective computing fusing physiological & acoustic signals. *IEEE Trans. Affect. Comput.*
- **Patel et al. (2026)**. Risk-constrained graph workflows for conversational mental health systems. *J. Med. Internet Res.*
- **Voultsiou, A. & Moussiades, L. (2026)**. Conversational agents in mental health: A systematic review. *Comput. Hum. Behav.*
- **Khubrani (2026)**. Multimodal stress detection using facial expression and speech dynamics. *Int. J. Hum.-Comput. Stud.*
- **Goodfellow, I. et al. (2013)**. Challenges in representation learning: FER-2013. *ICML Workshop*.
- **World Health Organization (2022)**. World Mental Health Report: Transforming Mental Health for All. Geneva.